

Vision-based Object Detection and Tracking of UAV with Unknown Camera Parameter

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Abstract—Vision-based object detection and tracking of Unmanned Aerial Vehicles (UAVs) has significant meaning for military or commercial applications in the fields of border patrol, target striking, criminal tracking, agricultural plant protection, and etc. To address the problem of UAV object detection and tracking of ground targets, this paper studies the vision-based object detection and target tracking control method for quad-rotor UAV with unknown camera parameter. Specifically, the YOLO v4 algorithm is utilized to achieve the onboard object detection of UAV and an optimized tracking controller based on LQR method is designed with camera parameters being unknown. Finally, a simulation environment is built based on the XTDrone simulation platform, and the effectiveness of the designed object detection and control algorithms is verified.

Keywords—quad-rotor UAV; object detection; YOLO v4; target tracking; optimized control

I. INTRODUCTION

In recent years, the small UAVs industry has developed rapidly as the small UAVs have advantages in flexibility, safety cost and operation convenience. Small UAVs can be equipped with different sensors and equipment for different scenarios. Currently, small drones are used in disaster rescue, target surveillance, tracking criminals, military reconnaissance and combat, as well as in commercial logistics and distribution.

Vision-based UAV target recognition and tracking has important applications in many civil or military scenarios. In the civilian sector, the UAV target recognition and tracking is widely used in disaster rescue, live sporting events and power maintenance. In military applications, the UAV target recognition and tracking can be used to significantly improve operational efficiency and to reduce soldier casualties. For example, when UAVs are deployed over enemy territory, they need to quickly and accurately detect, identify and monitor enemy targets through target identification and tracking technology, which can use

high-definition cameras to detect and monitor enemy combat vehicles, tanks, aircraft and other targets. The technology enables precise strike guidance for UAVs to collect information and provide accurate target strike guidance when soldiers are unable to obtain target coordinates directly, thereby increasing operational efficiency. In urban combat, UAV object detection and tracking technology can be used to distinguish between civilian buildings and military facilities as well as between military personnel and the general population, thus avoiding accidental damage to unrelated targets.

Accurate object detection and tracking is essentially important in many applications of UAVs. On the one hand, UAVs need to identify specific targets quickly and accurately in order to carry out their missions; on the other hand, UAVs need to track targets steadily in order to monitor the target persistently.

In 2016, Joseph Redmon et al. [1] proposed You Only Look Once (YOLO), an object detection algorithm which has been widely used owing to its fast detection speed. 2020, Alexey Bochkovskiy et al. [2] improved the previous research on the YOLO algorithm by combining various techniques, resulting in the YOLO v4 algorithm. The algorithm has the advantages of fast target detection, high accuracy and only requires a conventional GPU to train the dataset.

After the object detection is achieved through the YOLO v4 algorithm, the target is automatically framed in the image, after which the target is tracked by the UAV through subscribing to the target frame position information and designing the tracking control algorithm. Hyeonbeom Lee et al. pointed out in the literature [3] that basic linear controllers have been implemented to some extent for autonomous flight of multi-rotor UAVs. Where the PID controller is a universal controller, the linear quadratic regulator Linear Quadratic Regulator (LQR) is an internal mode control that optimises a quadratic cost function to obtain an optimal feedback gain matrix. For multi-rotor UAVs, feedback linearization methods such as

input-output linearization and state-space linearization have been proposed in the literature [4-6]. The automatic target tracking control of the UAV is achieved by using a negative feedback control system. In addition to PID control, an adaptive optimal control method based on LQR is applied for optimisation and a comparative analysis of the two tracking algorithms is carried out.

This paper addresses the problem of UAV detection and tracking of specific targets on the ground, and carries out research on the design of vision-based object detection and tracking control methods with unknown camera parameter for quadrotor UAVs, and builds a simulation environment for UAV tracking of unmanned vehicles on the ground based on the XTDrone simulation platform proposed by Kun Xiao et al. in [7] to verify the effectiveness of the designed target identification as well as control algorithms.

II. PROBLEM FORMULATION

Wherever Times is specified, Times Roman or Times New Roman may be used. If neither is available on your word processor, please use the font closest in appearance to Times. Avoid using bit-mapped fonts if possible. True-Type 1 or Open Type fonts are preferred. Please embed symbol fonts, as well, for math, etc.

A. Object Detection

An unmanned vehicle and a UAV equipped with a camera are present in the setup of the simulated experimental environment. The UAV is firstly controlled to fly and obtain images of the unmanned vehicle on the ground to collect sufficient dataset. Secondly, the collected dataset needs to be annotated, the target vehicle needs to be labeled using labeling software and the dataset needs to be classified into a training set as well as a testing set. Then, the YOLO v4 algorithm is used to train the dataset to obtain the corresponding network weight of the object detection algorithm. Finally, the obtained network weights are placed in the darknet_ros function package, the launch file for target recognition in the simulation environment is created and the online target recognition in the simulation environment is launched.

B. Control Model Establishment

After object detection is achieved by the YOLO v4 algorithm, the target is automatically framed in the image and then the target is tracked by the UAV through subscribing to the target frame position information and designing a tracking control algorithm. The control model is formulated in the following part.

1) *Pinhole Camera Model*: The pinhole camera model transforms the point $\tilde{P} = (X, Y, Z)$ in the real world into the point $\tilde{Q} = (x, y, z)$ in the image. The pinhole camera model is shown in Fig. 1, where X, Y, Z represent the 3D coordinates of the point $\tilde{P}$ in the real world coordinate system with respect to the main plane coordinate origin O and x, y, z represent the 3D coordinates of the projection point $\tilde{Q}$, which is the projection of $\tilde{P}$ in the image plane through the origin O .

The model of the pinhole camera is as follows

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{f}{Z} \begin{bmatrix} X \\ Y \end{bmatrix}, \quad (1)$$

where f represents the equivalent focal length of the camera. Then Equation (1) can be transformed into the following form

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, \quad (2)$$

where (x', y', z') is the homogeneous coordinate form of (x, y, z) .

Considering the difference of the origin and the direction of the axes between the world coordinate system and the camera coordinate system, the translation matrix T and the rotation matrix R are introduced. The matrix relation between the coordinates of the camera coordinate

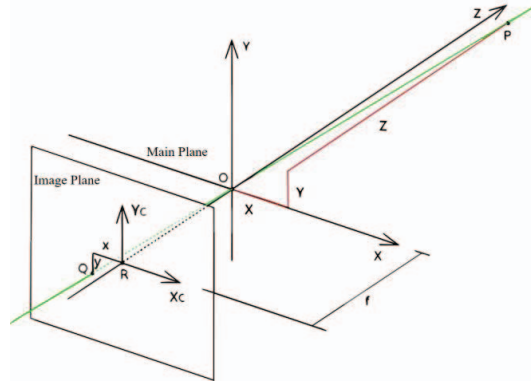


Figure 1. Pinhole camera model.

system $P = (x, y, 1)$ and the coordinates of the world coordinate system $Q = (X, Y, Z, 1)$ can be obtained as

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} R & T \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, \quad (3)$$

where $R = \sum_{i,j=1}^3 R_{ij}$, and $T = [T_x \ T_y \ T_z]^T$. The normalised pixel coordinates $L = (u, v, 1)$ are obtained by multiplying the coordinates P with the camera's internal parameters K , where the K matrix is as follows

$$K = \begin{bmatrix} f_x & \gamma & O_x \\ 0 & f_y & O_y \\ 0 & 0 & 1 \end{bmatrix}. \quad (4)$$

Expanding the matrix of internal and external camera parameters gives the following equation,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & \gamma & O_x \\ 0 & f_y & O_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (5)$$

Based on equation (3) and equation (5) above, it is deduced that

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & \gamma & O_x \\ 0 & f_y & O_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} & T_x \\ R_{21} & R_{22} & R_{23} & T_y \\ R_{31} & R_{32} & R_{33} & T_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}. \quad (6)$$

Then Equation (6) can be abbreviated as

$$\tilde{u} = K\tilde{x} = K \begin{bmatrix} R & T \end{bmatrix} \tilde{X}, \quad (7)$$

where K represents the internal parameter matrix of the camera, while R and T are the external parameters of the camera and represent the rotation and translation of the camera in the world coordinate system.

2) *Tracking Control Model*: Based on the pinhole camera model derived above, expanding equation (7) leads to the following equation,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = KR \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + KT. \quad (8)$$

Deriving both sides of equation (8) leads to

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ 0 \end{bmatrix} = KR \begin{bmatrix} \dot{X} \\ \dot{Y} \\ 0 \end{bmatrix} + K\dot{T}, \quad (9)$$

where X and Y here are the relative positions of the UAV and the target in the world coordinate system i.e., $X = p_x - p_{targetx}$, $Y = p_y - p_{targety}$, and T is the translation matrix, representing the translation of the camera in the world coordinate system, and its derivative is the velocity of the UAV in the camera coordinate system. Expanding equation (9) obtains the following equation,

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ 0 \end{bmatrix} = KR \begin{bmatrix} v_x - v_{targetx} \\ v_y - v_{targety} \\ 0 \end{bmatrix} + K \begin{bmatrix} v_x \\ v_y \\ 0 \end{bmatrix}, \quad (10)$$

where $v_{targetx}$ and $v_{targety}$ represent the velocity of the target in x and y directions respectively in the camera coordinate system. Then (10) can be rewritten as

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = K(R' + I) \begin{bmatrix} v_x \\ v_y \end{bmatrix} - KR' \begin{bmatrix} v_{targetx} \\ v_{targety} \end{bmatrix}, \quad (11)$$

where $R' = \sum_{i,j=1}^2 R_{ij}$ and I represents the unit matrix.

C. PID Controller Design

PID control is a well-known control method with a simple and adaptable structure. In the simulation environment, a PID control is designed as follows

$$\begin{aligned} v_x &= K_{px}e_x + K_{ix}e_{ix}\Delta t + K_{dx}\frac{e_x - e_{px}}{\Delta t}, \\ v_y &= K_{py}e_y + K_{iy}e_{iy}\Delta t + K_{dy}\frac{e_y - e_{py}}{\Delta t}, \end{aligned} \quad (12)$$

where K_p , K_i and K_d represent the proportional, integral and derivative coefficients respectively, e_x and e_y represent the position error of the target in the image at the x and y directions respectively, e_{px} and e_{py} represent

the previous error, e_{ix} and e_{iy} represent the cumulative error, and v_x and v_y represent the output velocity.

By adjusting the three coefficients K_p , K_i and K_d , the control law is able to achieve stable tracking of the UAV to the unmanned vehicle on the ground in the simulation environment.

D. LQR Optimal Controller Design

The LQR is a linear quadratic regulator that gives optimal control laws with linear feedback from the state. A computational adaptive optimal control for continuous-time linear systems with completely unknown dynamics method is presented in the literature [8]. Combined with this method the real-time target tracking is implemented in a designed simulation environment. The system is modelled as follows.

First define a continuous time linear system as

$$\dot{x} = Ax + Bu, \quad (13)$$

where x is the state of the system used for feedback control design and u is the controller input. Then assuming the system is stable, the design objective is to find the following linear optimal control law

$$u = -K_c x. \quad (14)$$

In the LQR controller, the output signal of the controller is calculated by using a feedback gain matrix K_c . The feedback gain matrix is determined by solving a cost function minimisation problem with the following performance metrics

$$J = \int_0^\infty (x^T Q x + u^T M u) dt. \quad (15)$$

The solution of the optimal control has been presented by Lewis et al. in [9] by solving the Riccati equation. The algorithm for solving the optimal feedback gain matrix K_c is shown in Algorithm 1. The inputs of the algorithm are matrices A , B and system weight matrices Q and M ; and the state weight matrix P and the state feedback matrix K_c are output.

Algorithm 1: Optimal Feedback Gain Matrix K_c

Input : State Space Matrix: A , B

Systematic Weighting Matrix: Q , M

Output: State Weight Matrix: P

State Feedback Gain Matrix: K_c

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1  $K_{c0} \leftarrow I$  % The initial value of  $K_{c0}$  set to unit matrix
2 for  $k \leftarrow 1$  to  $\infty$  do
3   while  $P^* \leq P_{k+1} \leq P_k$  do
4      $(A - BK_{ck})^T P_k + P_k (A - BK_{ck}) + Q +$ 
        $K_{ck}^T M K_{ck} \leftarrow 0;$ 
5      $K_{ck} \leftarrow M^{-1} B^T P_{k-1};$ 
6   end
7 end
8 return  $K_c$ 
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It can be observed in (11) that, when the target is static, $v_{targetx} = 0$, $v_{targety} = 0$, Equation (11) becomes

$$\begin{bmatrix} \dot{u} \\ \dot{v} \end{bmatrix} = K(R' + I) \begin{bmatrix} v_x \\ v_y \end{bmatrix}. \quad (16)$$

Based on equation (16), let the matrix $A=0$ and $B=K(R'+I)$ in equation (13). Then the UAV vision-based tracking control model can be transformed into the general model as follows

$$\dot{x} = Bu. \quad (17)$$

If the matrix B is known, it can be solved as in Algorithm 1 above. However, the matrix B is unknown due to the location of the camera's inner parameter matrix K and outer parameter matrix R' . The matrix B needs to be solved in a data-driven manner.

The data-driven solution to matrix B is as follows. First, given any control law $u=u(t)$, integrate both sides of the equation (17) over a time period T ,

$$\int_{t_0}^{t_0+T} \dot{x} d\tau = \int_{t_0}^{t_0+T} Bu d\tau. \quad (18)$$

Expanding equation (18) gives

$$x(t_0+T) - x(t_0) = B \int_{t_0}^{t_0+T} u d\tau, \quad (19)$$

assuming that the left-hand side of equation (19) is written as $X(t_0)$ and the right-hand side as $\phi(t_0)$. Assuming that samples are collected for N time periods, we can obtain that

$$\begin{bmatrix} X(t_0) & X(t_1) & \dots & X(t_N) \end{bmatrix} = B \begin{bmatrix} \phi(t_0) & \phi(t_1) & \dots & \phi(t_N) \end{bmatrix}, \quad (20)$$

the parameters in the x and y directions of the control model are integrated, and the left-hand side of equation (20) is noted as $\omega_{2 \times N}$ and the right-hand side as $Y_{2 \times N}$, resulting in the following calculation of the parameter matrix B .

$$\begin{aligned} \omega_{2 \times N} &= B_{2 \times 2} Y_{2 \times N}, \\ \omega Y^T &= B Y Y^T, \\ B &= \omega Y^T (Y Y^T)^{-1}. \end{aligned} \quad (21)$$

The optimal feedback gain matrix K_c can be obtained by solving the parameter matrix B from equations (18 ~ 21) above into Algorithm 1, thus obtaining the linear optimal control law of equation (14).

III. SIMULATION EXPERIMENTS

The simulation experiments are mainly based on the open source XTDrone platform, which is a customisable multi-rotor UAV simulation platform based on ROS, Gazebo and PX4, and integrates dynamics models, sensors, control algorithms, state estimation algorithms and 3D scenarios to facilitate the design and deployment of the algorithms.

A. Simulation Scene Construction

The first step is to construct the simulation scene, the envisaged simulation scene is an unmanned vehicle and a UAV, build the world scene by creating a world file, then set the parameters in the camera's sdf file, then create the corresponding launch file, and launch the simulation scene of the launch file as shown in the Fig. 2. In this scene the camera is oriented 90° downwards by modifying the camera parameter pose frame.

B. Collecting Data Sets

Control the UAV to fly at different altitudes and open image_view to collect the data sets. Collect a sufficient number of images of unmanned vehicles on the ground with different angles and sizes. In subsequent simulation experiments it was found that online recognition would identify the shadow of the ground-based UAV as the target, therefore, 20 images of ground-based UAV projections were added to the original 200 image data sets.

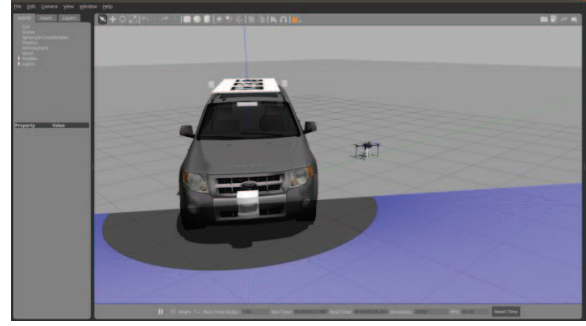


Figure 2. Simulation scenes constructed in Gazebo.

C. Data Sets Tagging And Training

Based on the data sets obtained above, the labeling tool was used to label the above data sets and classify the data sets into training and testing sets. An example of the labelling process is shown in Fig. 3.

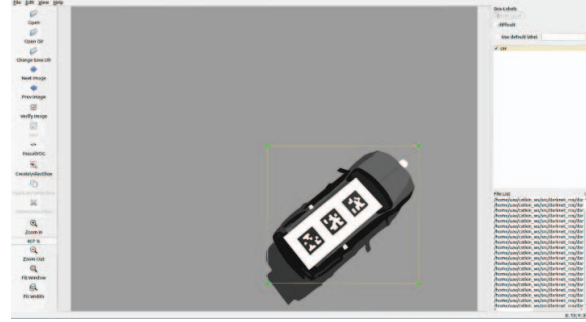


Figure 3. Example of data sets tagging.

After the data sets are fully labelled and classified, download the pre-training weights of YOLO v4 and start the training of the data sets.

D. Online Object Detection In The Simulation Environment

The darknet_ros function package implements the target detection function in the ROS robotics system and is compiled to facilitate the training of subsequent object detection data sets. The network weights obtained from the training are first placed in the corresponding darknet_ros directory. Then the launch file corresponding to the object detection task is created, and YOLO v4 is launched in the simulation environment, thus enabling online object detection in the simulation environment. Online object detection is shown in Fig. 4.

E. Target Tracking Control

After implementing online target recognition in the simulation environment, the target is tracked in real time based on the position information obtained by feedback.

The implementation flow of the target tracking control algorithm is described as follows. First, initiate the node, then create publishers and subscribers to subscribe to the target position. The position error is calculated based on

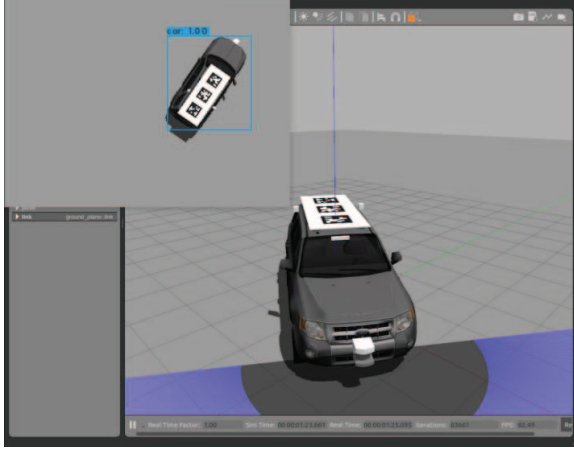


Figure 4. Online object detection.

the obtained target position and the expected centre position of the image. PID control or LQR optimisation control is then performed to obtain the output velocity.

IV. MAIN RESULTS

A. YOLO v4 Object Detection Results

The data sets collected in Section III-B has around 220 images, 200 of which are images of unmanned vehicles on the ground and 20 of which are images of UAV ground shadows. The results of the data sets training are shown in Fig. 5 and TABLE 1, where mAP and FPS are the abbreviation of Average Precision and Frame Per Second.

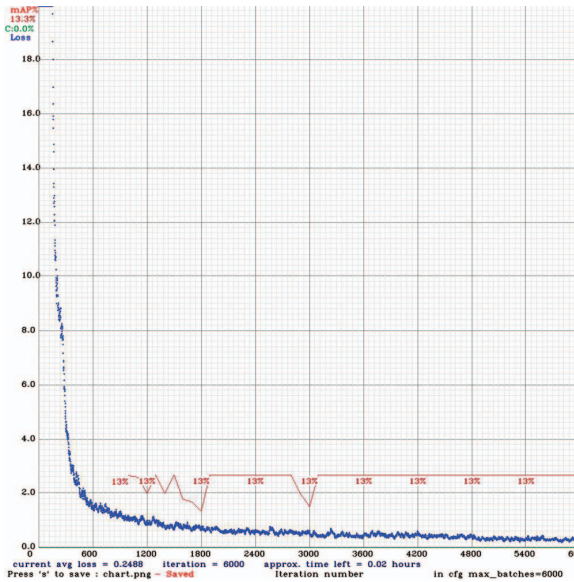


Figure 5. YOLO v4 object detection result.

TABLE I. YOLO v4 OBJECT DETECTION RESULTS

Indicators	mAP	Precision	Average IoU	FPS
Data	13.33%	1.00	91.54%	76.66

The average frame rate obtained from the training is high enough for online detection. But the data obtained based on YOLO v4 training, where the average accuracy was only 13.33%, may be due to the small data sets and the small number of data set classes, hence the lack of generalisation of the resulting detectors.

B. Tracking Stationary Target Results

The tracking effects of the PID tracking control and the LQR optimised control are verified in the simulation environment respectively.

Firstly, the UAV was controlled to track a stationary target and the tracking effects under both control algorithms are shown in Fig. 6 and Fig. 7. Fig. 6 shows the comparison of the position error of the two control algorithms for tracking a stationary target. Fig. 7 shows the comparison of the velocity control of two control algorithms tracking a stationary target.

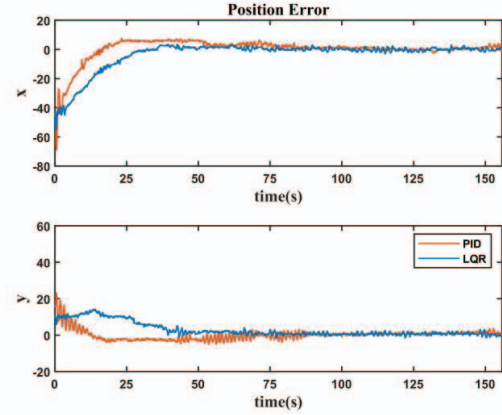


Figure 6. Tracking stationary target position error.

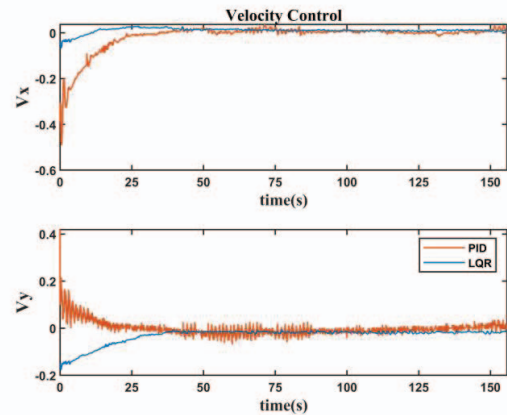


Figure 7. Tracking stationary target velocity control.

From the simulation results, it can be observed that both controllers achieve stable tracking of the stationary target, with both achieving a relatively stable state after 60s. The stability and robustness of the LQR optimised

controller for target tracking is shown to be superior to that of the PID controller.

V. CONCLUSIONS

The paper investigates vision-based object detection and tracking methods for UAVs, with the main contents as follows.

- Designed an online object detection algorithm based on YOLO v4.
- Designed a PID-based target tracking algorithm and an LQR control law with unknown camera parameters.
- Verified the algorithms based on the XTDrone simulation platform.

The work in the paper is primarily based on a simulation platform that can be deployed on a physical UAV in the future. On the basis of this research, interference can be actively applied to the simulation environment to simulate realistic flight.

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